

SOC 690S: Machine Learning in Causal Inference

Week 10: Sharp Regression Discontinuity Design

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Week	Date	Topic	Problem sets	
			Assign	Due
1	Aug 26	Introduction: Motivation and Linear Regression		
2	Sep 2	Foundation: ML-based Regression Regularization		
3	Sep 9	Foundation: Non-Linear Machine Learning	1	
4	Sep 16	Foundation: Double Machine Learning		
5	Sep 23	Foundation: Directed Acyclic Graph (DAG)	2	1
6	Sep 29	Core: Matching, Propensity, Weighting, and Beyond		
7	Oct 7	Core: Instrumental Variable Estimation	3	2
8	Oct 14	<i>Fall break</i>		
9	Oct 21	In-class midterm		
10	Oct 28	Core: Regression Discontinuity Design		
11	Nov 4	Core: Panel Data and Difference-in-Difference	4	3
12	Nov 11	Advanced: Heterogeneous Treatment Effect		
13	Nov 18 [†]	Advanced: Unstructured Data Feature Engineering	5	4
14	Nov 25	Advanced: Causal Reasoning in Machine Learning		
	Dec 2	Take-home final		5

[†] Justin Lucas Sola, Assistant Professor of Sociology and Data Science at UNC, will join us to give a guest talk on heterogeneous treatment effects and their applications.

Sharp Regression Discontinuity (RD): Setup

- In sharp RD, treatment status is a deterministic and discontinuous function a covariate x_i
- Suppose, for example, that

$$D_i = \begin{cases} 1 & \text{if } x_i \geq c \\ 0 & \text{if } x_i < c \end{cases}$$

where c is a known threshold or cutoff

- The assignment mechanism is a deterministic function of x_i because once we know x_i we know D_i
- Treatment is a discontinuous function of x_i because no matter how close x_i gets to c , treatment is unchanged until $x_i = c$

Sharp Regression Discontinuity (RD): Assumptions

- There are two necessary assumptions for RD identification
- 1. Local regularity, or continuity of potential outcomes

$$\lim_{x \downarrow c} E[Y_i(0) | X_i = x] = \lim_{x \uparrow c} E[Y_i(0) | X_i = x]$$

$$\lim_{x \downarrow c} E[Y_i(1) | x_i = x] = \lim_{x \uparrow c} E[Y_i(1) | x_i = x]$$

- It implies that the potential outcomes evolve smoothly in X_i ; there are no unobserved characteristics that discontinuously change at c other than X_i
- 2. No manipulation of the running variable: individuals cannot sort or manipulate X_i around c ; the density $f(X_i)$ is continuous at the cutoff

$$\lim_{x \downarrow c} f(x_i) = \lim_{x \uparrow c} f(x_i)$$

- Both can be tested using McCrary (2008) (in `rdd:DCdensity()` in R)

Sharp Regression Discontinuity (RD): Setup

The parameter of interest in RDD is the local ATE at the cutoff value c

$$\tau = E[Y_i(1) - Y_i(0) | X_i = c]$$

Under local regularity

$$\begin{aligned}\tau &= \lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x] \\ &= g_+(c) - g_-(c)\end{aligned}$$

when we express CEF by $g(\cdot)$

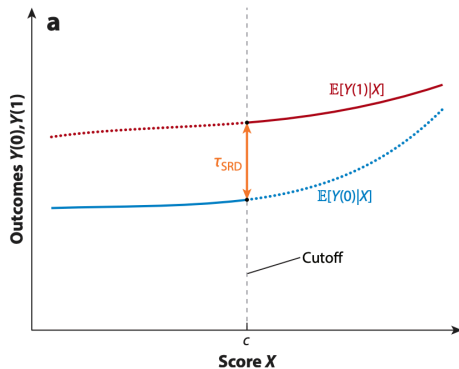


Figure: Sharp RDD setup

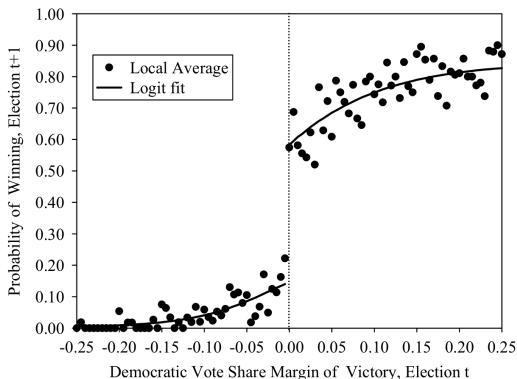
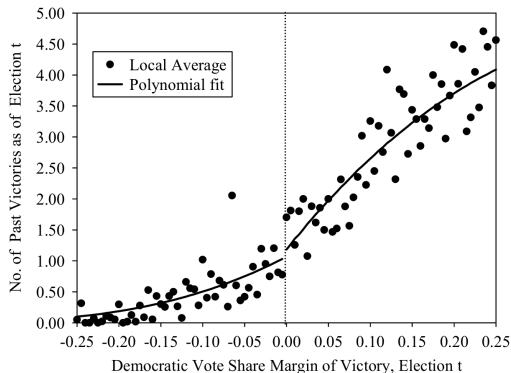
Sharp Regression Discontinuity (RD): Example

- One question in social science is whether receiving academic recognition in early career has long-term positive effects
- Does receiving finalist status or eligibility for a merit-based scholarship affects students' later educational outcomes—such as college attendance, college quality, or academic achievement
- PSAT score determines the eligibility for National Merit Scholarship
- Students just above and below the PSAT threshold c are nearly identical in ability; any discontinuity in $E[Y_i | X_i]$ at c is interpreted as the causal effect of eligibility (Campbell 1969)

Sharp Regression Discontinuity (RD): Lee (2008)

- Lee (2008) examined the causal effect of incumbency on electoral success in U.S. House races
- Focused on elections that are extremely close, where two candidates—typically a Democrat and a Republican—receive nearly the same share of the vote
- In these very tight contests, the outcome is determined by essentially random factors such as last-minute events, turnout fluctuations, or counting noise
- Candidates who barely win and those who barely lose are, on average, comparable in ability, popularity, and campaign resources

Regression Discontinuity (RD): Lee (2008)



Regression Discontinuity (RD): Lei and Zhou (2022)

- Lei and Zhou (2022) explored why Chinese local governments invest heavily in long-term infrastructure projects—especially in situations where a local leader's tenure is short and the project pay-offs extend far into the future
- Local officials can still gain private political returns by initiating high-visibility infrastructure projects that elevate the economic performance of their jurisdiction
- To be eligible for creating a subway system, a city must satisfy four central government requirements (1) the city's annual fiscal revenue exceeds 10 billion, (2) the city's GDP reaches at least 100 billion, (3) the city's population exceeds 3 million people, and (4) more than 30,000 people per hour are expected to use a subway line
- While cities satisfying these four requirements are not guaranteed to have subways, those not satisfying them are not eligible

Sharp Regression Discontinuity (RD): Local Fit

- An important feature of RD is that there is *no* value of X_i at which we get to observe both treatment and control observations
- Unlike full covariate matching strategies, which are based on treatment-control comparisons conditional on covariate values where there is some overlap, the validity of RD turns on our willingness to extrapolate across covariate values, at least in a neighborhood of the discontinuity
- The functional form of the regression becomes essential for estimation

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- The functional form of the regression becomes essential for estimation
- The most common functional form of Sharp RD is a (local) linear regression

$$Y_i = a_+ + b_+(x_i - c) + e_i \quad \text{for } x_i \geq c$$

$$Y_i = a_- + b_-(x_i - c) + e_i \quad \text{for } x_i < c$$

$$\hat{\tau} = a_+ - a_-$$

Bandwidth Selection Intuition

- Given the estimated ATE is local to the *cutoff*, ideally the bandwidth h used in local linear regression should be reasonably small
- To avoid introducing large *biases* from including observations further away from the cutoff that may be systematically different from observations at the cutoff in unobserved ways (therefore distorting the prediction of a_+ and a_-)
- But it cannot be too small, as otherwise very few observations will be used; estimation uncertainty (variance) would be large
- This is the classic bias–variance tradeoff

Local Kernel Weight

- Besides tuning the bandwidth, a second (and complementary) way to reduce bias from distant observations is to apply *local kernel weights*
- Replace raw weights by $K(u)$ so observations farther from the cutoff get less influence, where $u = (x - c)/h$
- Common kernels are

$$\begin{cases} K(u) = (1 - |u|) \cdot \mathbb{1}(|u| \leq 1) & \text{triangular} \\ K(u) = \frac{3}{4}(1 - u^2) \cdot \mathbb{1}(|u| \leq 1) & \text{Epanechnikov} \\ K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) & \text{Gaussian} \end{cases}$$

- *Triangular* kernel is often recommended in RDD practice; it places greater weight on observations closest to the cutoff and reduces boundary bias (the default in many RDD packages)
- Choice of kernel is much less important than choice of bandwidth

Local Regression Form

- For the same reasons, high-order polynomials are not recommended as they may introduce significant biases in extrapolation and prediction at cutoff c if the true DGP does *not* have high-order terms

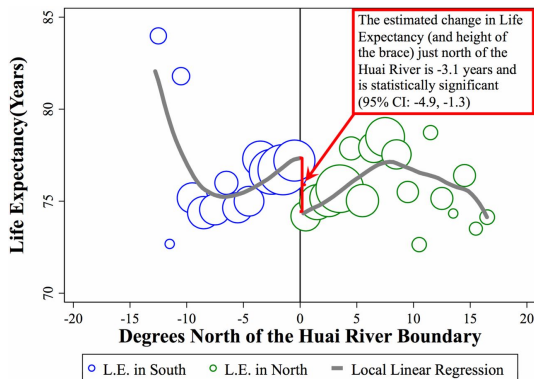


Figure: Ebenstein et al. (2017)

Local Regression Form

- If there are sufficient number of observations around the cutoff c , a more natural estimand is a local nonparametric mean difference

$$E[Y_i(0) \mid X_i = c] \simeq E[Y_i \mid c - \Delta < X_i < c]$$

$$E[Y_i(1) \mid X_i = c] \simeq E[Y_i \mid c \leq X_i < c + \Delta]$$

$$\lim_{\Delta \rightarrow 0} E[Y_i \mid c \leq X_i < c + \Delta] - E[Y_i \mid c - \Delta < X_i < c] = E[Y_i(1) - Y_i(0) \mid X_i = c]$$

- The bandwidth h is chosen as the largest local window where local randomization assumption (covariate balance) is met (Cattaneo, Frandsen, and Titiunik 2015)

Bias and Variance Tradeoff in Bandwidth Selection

- Recall that we define right and left limits

$$g_+(c) = \lim_{x \downarrow c} g(x), \quad g_-(c) = \lim_{x \uparrow c} g(x)$$

- And we estimate the treatment effect at the cutoff

$$\tau = g_+(c) - g_-(c)$$

by separate *one-sided* local linear regressions using bandwidth h

- Focusing on the right hand side of the cutoff with a kernel function $K(\cdot)$

$$(\hat{a}_+, \hat{b}_+) = \arg \min_{a, b} \sum_{i; X_i > c} K\left(\frac{X_i - c}{h}\right) [Y_i - a - b(x_i - c)]^2$$

Bias and Variance Tradeoff in Bandwidth Selection

- The mean squared error (MSE) of $\hat{a}_+$ can be decomposed as

$$E \left[(\hat{a}_+ - g_+(c))^2 \right] = (E[\hat{a}_+] - g_+(c))^2 + V(\hat{a}_+)$$

- The optimal bandwidth h minimizes the MSE

Bias and Variance Tradeoff in Bandwidth Selection

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- The optimal bandwidth h minimizes the MSE
- To make notations a bit simpler, define

$$u_i = \frac{X_i - c}{h} \quad \text{such that } u_i \in [0, 1]$$

- Expand Y_i at cutoff c using Taylor expansion

$$\begin{aligned} Y_i &= g_+(X_i) + e_i \\ &= g_+(c) + g'_+(c)(X_i - c) + \frac{1}{2}g''_+(c)(X_i - c)^2 + O_3 + e_i \\ &= g_+(c) + g'_+(c)hu_i + \frac{1}{2}g''_+(c)h^2u_i^2 + e_i \end{aligned}$$

Bias and Variance Tradeoff in Bandwidth Selection

- Solving the *population* normal equation for the intercept a_+

$$\begin{aligned} 0 &= E [K(u_i)(Y_i - a_+ - b_+hu_i)] \\ &= E \left[K(u_i) \left(g_+(c) + g'_+(c)hu_i + \frac{1}{2}g''_+(c)h^2u_i^2 - a_+ - b_+hu_i \right) \right] \\ &= \int_0^1 K(u) \left(g_+(c) + g'_+(c)hu + \frac{1}{2}g''_+(c)h^2u^2 - a_+ - b_+hu \right) du \\ &= (g_+(c) - a_+) \int_0^1 K(u)du + (g'_+(c) - b_+)h \int_0^1 uK(u)du + \\ &\quad \frac{1}{2}g''_+(c)h^2 \int_0^1 u^2K(u)du \end{aligned}$$

Bias and Variance Tradeoff in Bandwidth Selection

- Define the kernel moments

$$m_j = \int_0^1 u^j K(u) du, \quad j = 0, 1, 2$$

$$\begin{aligned} 0 &= (g_+(c) - a_+)m_0 + (g'_+(c) - b_+)hm_1 + \frac{1}{2}g''_+(c)h^2m_2 \\ &= (g_+(c) - a_+) + (g'_+(c) - b_+)h\frac{m_1}{m_0} + \frac{1}{2}g''_+(c)h^2\frac{m_2}{m_0} \\ &= (g_+(c) - a_+) + (g'_+(c) - b_+)h\mu_1 + \frac{1}{2}g''_+(c)h^2\mu_2 \end{aligned}$$

where

$$\mu_j = \frac{m_j}{m_0}$$

Bias and Variance Tradeoff in Bandwidth Selection

- Define the *bias* of the estimated a_+

$$\Delta(\hat{a}_+) = \hat{a}_+ - g_+(c)$$

- We can similarly solve the *population* normal equation for the slope b_+ , with

$$\Delta(\hat{b}_+) = h \left(\hat{b}_+ - g_+(c) \right)$$

- Solving the two normal equations

$$\begin{aligned}\Delta(\hat{a}_+) + \mu_1 \Delta(\hat{b}_+) &= -\frac{1}{2} h^2 g_+''(c) \mu_2 \\ \mu_1 \Delta(\hat{a}_+) + \mu_2 \Delta(\hat{b}_+) &= -\frac{1}{2} h^2 g_+''(c) \mu_3\end{aligned}$$

Bias and Variance Tradeoff in Bandwidth Selection

- Simplifying the two normal equations, we will derive

$$\begin{aligned}\Delta(\hat{a}_+) &= \frac{h^2}{2} g_+''(c) \frac{\mu_1 \mu_3 - \mu_2^2}{\mu_2 - \mu_1^2} \\ &= \frac{h^2}{2} g_+''(c) A_k\end{aligned}$$

Bias and Variance Tradeoff in Bandwidth Selection

- In more compact matrix notation

$$\Delta(\hat{a}_+) = v_1' (E[K(u_i)X_iX_i'])^{-1} E[K(u_i)X_ie_i]$$

$$V(\hat{a}_+) = v_1' (E[K(u_i)X_iX_i'])^{-1} V(E[K(u_i)X_ie_i]) (E[K(u_i)X_iX_i'])^{-1} v_1$$

where $v_1 = (1, 0)$

- The intuition is from the standard OLS, where

$$\hat{\beta} - \beta = (X'X)^{-1}X'e$$

Bias and Variance Tradeoff in Bandwidth Selection

- Suppose u_i has density approximately $f(c)$ near c

$$E[K(u_i)X_iX_i'] = f(c) \int_0^1 K(u) \begin{pmatrix} 1 \\ u \end{pmatrix} (1, u) du = f(c) \begin{pmatrix} m_0 & m_1 \\ m_1 & m_2 \end{pmatrix}$$

- And the “meat”

$$\begin{aligned} V(E[K(u_i)X_ie_i]) &= E[K(u_i)^2X_iX_i'e_i^2] \\ &= \sigma_+^2 f(c) \int_0^1 K(u)^2 \begin{pmatrix} 1 \\ u \end{pmatrix} (1, u) du \\ &= \sigma_+^2 f(c) \begin{pmatrix} r_0 & r_1 \\ r_1 & r_2 \end{pmatrix} \end{aligned}$$

where

$$\sigma_+^2 = V(e_i | X_i) \text{ and } r_j = \int_0^1 u^j K(u)^2 du$$

Bias and Variance Tradeoff in Bandwidth Selection

- Simplifying terms, we will eventually derive defined as

$$\begin{aligned} V(\hat{a}_+) &= \frac{\sigma_+^2}{nhf(c)} \frac{m_2^2 r_0 - 2m_1 m_2 r_1 + m_1^2 r^2}{(m_0 m_2 - m_1^2)^2} \\ &= \frac{\sigma_+^2}{nhf(c)} B_K \end{aligned}$$

where $nhf(c)$ is the approximate effective size in window $[c, c + h]$

The Imbens–Kalyanaraman Bandwidth

- For the estimated treatment effect $\hat{\tau}$, the sum of mean squared error can be decomposed as the sum of the square of bias and variance

$$\Delta(\hat{\tau})^2 + V(\hat{\tau}) = \frac{A_K^2}{4} (g_+''(c) - g_-''(c))^2 h^4 + \frac{B_K}{nhf(c)} (\sigma_+^2 - \sigma_-^2)$$

- Solving FOC *w.r.t.* h to minimize the sum and to strike a balance between variance and bias, we have

$$h_{IK} = \left[\frac{B_K}{A_K^2} \frac{\sigma_+^2 + \sigma_-^2}{nf(c) (g_+''(c) - g_-''(c))^2} \right]^{1/5}$$

Covariates Controls in Sharp RD

Motivations of Including Covariates

- The most important benefits of including covariates is to improve estimation efficiency
- As in RCTs, using covariates can increase efficiency and improve power
- The intuition is that the variance of the estimand is a function of σ^2 , the variance of the error term unexplained by X_i (see a formal derivation of the quantity on page 475 in Chapter 17 of CML)
- Importantly, the estimation bias is *not* a function of covariates

Including Covariates in Sharp RD

- In the regular RDD without covariates, the compact approach to write the estimand of ATE is

$$\hat{\tau}(h) = v_2' \arg \min_{\theta \in \mathbb{R}^4} \sum_{i=1}^n K_h(X_i - c) (Y_i - \theta' V_i)^2$$

- $V_i = (1, D_i, (X_i - c), D_i(X_i - c))'$
- $K_h(x)$ is kernel weights around c
- h = bandwidth
- $v_2 = (0, 1, 0, 0)'$ selects coefficient on D_i

Including Covariates in Sharp RD

- With covariates, the compact approach to write the estimand of ATE is

$$\hat{\tau}(h) = v_2' \arg \min_{\theta, \gamma \in \mathbb{R}^{(4+p)}} \sum_{i=1}^n K_h(X_i - c) (Y_i - \theta' V_i - \gamma' Z_i)^2$$

- This will be equivalent to the residualized version

$$\hat{\tau}(h) = v_2' \arg \min_{\theta \in \mathbb{R}^4} \sum_{i=1}^n K_h(X_i - c) (\tilde{Y}_i - \theta' V_i)^2$$

- where

$$\tilde{Y}_i = Y_i - Z_i' \hat{\gamma}$$

$$\hat{\gamma} = \arg \min_{\gamma} \sum_{i=1}^n K_h(X_i - c) (Y_i - \gamma' Z_i)^2$$

Including Covariates in Sharp RD

- As before, the estimation of γ does not have to be linear or in low dimension
- One may flexibly estimate the nuisance function $\eta_0(Z_i) = E[Y_i | Z_i]$ using ML models and define the covariate-adjusted ATE as

$$\tau = \lim_{x \downarrow c} E[Y_i - \eta_0(Z_i) | X_i = x] - \lim_{x \uparrow c} E[Y_i - \eta_0(Z_i) | X_i = x]$$